System size and energy dependence of near-side jet-like correlations and the ridge at RHIC

M. M. Aggarwal,[29] Z. Ahammed,[21] A. V. Alakhverdyants,[17] I. Alekseev ,[15] J. Alford,[18] B. D. Anderson,[18] C. D. Anson,[27] D. Arkhipkin,[2] G. S. Averichev,[17] J. Balewski,[22] D. R. Beavis,[2] R. Bellwied,[49] M. J. Betancourt,[22] R. R. Betts,[7] A. Bhasin,[16] A. K. Bhati,[29] H. Bichsel,[48] J. Bielcik,[9] J. Bielcikova,[10] B. Biritz,[5] L. C. Bland,[2] W. Borowski,[40] J. Bouchet,[18] E. Braidot,[26] A. V. Brandin,[25] A. Bridgeman,[1] S. G. Brovko,[4] E. Bruna,[51] S. Bueltmann,[28] I. Bunzarov,[17] T. P. Burton,[2] X. Z. Cai,[39] H. Caines,[51] M. Calderón de la Barca Sánchez,[4] D. Cebra,[4] R. Cendejas,[5] M. C. Cervantes,[41] Z. Chajecki,[27] P. Chaloupka,[10] S. Chattopadhyay,[46] H. F. Chen,[37] J. H. Chen,[39] J. Y. Chen,[50] J. Cheng,[43] M. Cherney,[8] A. Chikanian,[51] K. E. Choi,[33] W. Christie,[2] P. Chung,[10] M. J. M. Codrington,[41] R. Corliss,[22] J. G. Cramer,[48] H. J. Crawford,[3] S. Dash,[12] A. Davila Leyva,[42] L. C. De Silva,[49] R. R. Debbe,[2] T. G. Dedovich,[17] A. A. Derevschikov,[31] R. Derradi de Souza,[6] L. Didenko,[2] P. Djawotho,[41] S. M. Dogra,[16] X. Dong,[21] J. L. Drachenberg,[41] J. E. Draper,[4] J. C. Dunlop,[2] M. R. Dutta Mazumdar,[46] L. G. Efimov,[17] M. Elnimr,[49] J. Engelage,[3] G. Eppley,[35] M. Estienne,[40] L. Eun,[30] O. Evdokimov,[7] R. Fatemi,[19] J. Fedorisin,[17] R. G. Fersch,[19] E. Finch,[51] V. Fine,[2] Y. Fisyak,[2] C. A. Gagliardi,[41] D. R. Gangadharan,[5] A. Geromitsos,[40] F. Geurts,[35] P. Ghosh,[46] Y. N. Gorbunov,[8] A. Gordon,[2] O. Grebenyuk,[21] D. Grosnick,[45] S. M. Guertin,[5] A. Gupta,[16] W. Guryn,[2] B. Haag,[4] O. Hajkova,[9] A. Hamed,[41] L-X. Han,[39] J. W. Harris,[51] J. P. Hays-Wehle,[22] M. Heinz,[51] S. Heppelmann,[30] A. Hirsch,[32] E. Hjort,[21] G. W. Hoffmann,[42] D. J. Hofman,[7] B. Huang,[37] H. Z. Huang,[5] T. J. Humanic,[27] L. Huo,[41] G. Igo,[5] P. Jacobs,[21] W. W. Jacobs,[14] C. Jena,[12] F. Jin,[39] J. Joseph,[18] E. G. Judd,[3] S. Kabana,[40] K. Kang,[43] J. Kapitan,[10] K. Kauder,[7] D. Keane,[18] A. Kechechyan,[17] D. Kettler,[48] D. P. Kikola,[21] J. Kiryluk,[21] A. Kisiel,[47] V. Kizka,[17] S. R. Klein,[21] A. G. Knospe,[51] D. D. Koetke,[45] T. Kollegger,[11] J. Konzer,[32] I. Koralt,[28] L. Koroleva,[15] W. Korsch,[19] L. Kotchenda,[25] V. Kouchpil,[10] P. Kravtsov,[25] K. Krueger,[1] M. Krus,[9] L. Kumar,[18] P. Kurnadi,[5] M. A. C. Lamont,[2] J. M. Landgraf,[2] S. LaPointe,[49] J. Lauret,[2] A. Lebedev,[2] R. Lednicky,[17] J. H. Lee,[2] W. Leight,[22] M. J. LeVine,[2] C. Li,[37] L. Li,[42] N. Li,[50] W. Li,[39] X. Li,[32] X. Li,[38] Y. Li,[43] Z. M. Li,[50] M. A. Lisa,[27] F. Liu,[50] H. Liu,[4] J. Liu,[35] T. Ljubicic,[2] W. J. Llope,[35] R. S. Longacre,[2] W. A. Love,[2] Y. Lu,[37] E. V. Lukashov,[25] X. Luo,[37] G. L. Ma,[39] Y. G. Ma,[39] D. P. Mahapatra,[12] R. Majka,[51] O. I. Mall,[4] L. K. Mangotra,[16] R. Manweiler,[45] S. Margetis,[18] C. Markert,[42] H. Masui,[21] H. S. Matis,[21] Yu. A. Matulenko,[31] D. McDonald,[35] T. S. McShane,[8] A. Meschanin,[31] R. Milner,[22] N. G. Minaev,[31] S. Mioduszewski,[41] A. Mischke,[26] M. K. Mitrovski,[11] B. Mohanty,[46] M. M. Mondal,[46] B. Morozov,[15] D. A. Morozov,[31] M. G. Munhoz,[36] M. Naglis,[21] B. K. Nandi,[13] T. K. Nayak,[46] P. K. Netrakanti,[32] M. J. Ng,[3] L. V. Nogach,[31] S. B. Nurushev,[31] G. Odyniec,[21] A. Ogawa,[2] Oh,[33] Ohlson,[51] V. Okorokov,[25] E. W. Oldag,[42] D. Olson,[21] M. Pachr,[9] B. S. Page,[14] S. K. Pal,[46] Y. Pandit,[18] Y. Panebratsev,[17] T. Pawlak,[47] H. Pei,[7] T. Peitzmann,[26] C. Perkins,[3] W. Peryt,[47] S. C. Phatak,[12] P. Pile,[2] M. Planinic,[52] M. A. Ploskon,[21] J. Pluta,[47] D. Plyku,[28] N. Poljak,[52] A. M. Poskanzer,[21] B. V. K. S. Potukuchi,[16] C. B. Powell,[21] D. Prindle,[48] C. Pruneau,[49] N. K. Pruthi,[29] P. R. Pujahari,[13] J. Putschke,[51] H. Qiu,[20] R. Raniwala,[34] S. Raniwala,[34] R. L. Ray,[42] R. Redwine,[22] R. Reed,[4] H. G. Ritter,[21] J. B. Roberts,[35] O. V. Rogachevskiy,[17] J. L. Romero,[4] A. Rose,[21] L. Ruan,[2] J. Rusnak,[10] S. Sakai,[21] I. Sakrejda,[21] T. Sakuma,[22] S. Salur,[4] J. Sandweiss,[51] E. Sangaline,[4] J. Schambach,[42] R. P. Scharenberg,[32] A. M. Schmah,[21] N. Schmitz,[23] T. R. Schuster,[11] J. Seele,[22] J. Seger,[8] I. Selyuzhenkov,[14] P. Seyboth,[23] E. Shahaliev,[17] M. Shao,[37] M. Sharma,[49] S. S. Shi,[50] E. P. Sichtermann,[21] F. Simon,[23] R. N. Singaraju,[46] M. J. Skoby,[32] N. Smirnov,[51] P. Sorensen,[2] H. M. Spinka,[1] B. Srivastava,[32] T. D. S. Stanislaus,[45] D. Staszak,[5] S. G. Steadman,[22] J. R. Stevens,[14] R. Stock,[11] M. Strikhanov,[25] B. Stringfellow,[32] A. A. P. Suaide,[36] M. C. Suarez,[7] N. L. Subba,[18] M. Sumbera,[10] X. M. Sun,[21] Y. Sun,[37] Z. Sun,[20] B. Surrow,[22] D. N. Svirida,[15] T. J. M. Symons,[21] A. Szanto de Toledo,[36] J. Takahashi,[6] A. H. Tang,[2] Z. Tang,[37] L. H. Tarini,[49] T. Tarnowsky,[24] D. Thein,[42] J. H. Thomas,[21] J. Tian,[39] A. R. Timmins,[49] D. Tlusty,[10] M. Tokarev,[17] T. A. Trainor,[48] V. N. Tram,[21] S. Trentalange,[5] R. E. Tribble,[41] Tribedy,[46] O. D. Tsai,[5] T. Ullrich,[2] D. G. Underwood,[1] G. Van Buren,[2] G. van Nieuwenhuizen,[22] J. A. Vanfossen, Jr.,[18] R. Varma,[13] G. M. S. Vasconcelos,[6] A. N. Vasiliev,[31] F. Videbæk,[2] Y. P. Viyogi,[46] S. Vokal,[17] S. A. Voloshin,[49] M. Wada,[42] M. Walker,[22] F. Wang,[32] G. Wang,[5] H. Wang,[24] J. S. Wang,[20] Q. Wang,[32] X. L. Wang,[37] Y. Wang,[43] G. Webb,[19] J. C. Webb,[2] G. D. Westfall,[24] C. Whitten Jr.,[5] H. Wieman,[21] S. W. Wissink,[14] R. Witt,[44] W. Witzke,[19] Y. F. Wu,[50] W. Xie,[32] H. Xu,[20] N. Xu,[21] Q. H. Xu,[38] W. Xu,[5] Y. Xu,[37] Z. Xu,[2] L. Xue,[39] Y. Yang,[20] P. Yepes,[35] K. Yip,[2] I-K. Yoo,[33] Q. Yue,[43] M. Zawisza,[47] H. Zbroszczyk,[47] W. Zhan,[20] J. B. Zhang,[50] S. Zhang,[39] W. M. Zhang,[18] X. P. Zhang,[43] Y. Zhang,[21] Z. P. Zhang,[37] J. Zhao,[39] C. Zhong,[39] W. Zhou,[38] X. Zhu,[43] Y. H. Zhu,[39] R. Zoulkarneev,[17] and Y. Zoulkarneeva[17]

(STAR Collaboration)

[1]Argonne National Laboratory, Argonne, Illinois 60439, USA
[2]Brookhaven National Laboratory, Upton, New York 11973, USA
[3]University of California, Berkeley, California 94720, USA
[4]University of California, Davis, California 95616, USA
[5]University of California, Los Angeles, California 90095, USA
[6]Universidade Estadual de Campinas, Sao Paulo, Brazil
[7]University of Illinois at Chicago, Chicago, Illinois 60607, USA
[8]Creighton University, Omaha, Nebraska 68178, USA
[9]Czech Technical University in Prague, FNSPE, Prague, 115 19, Czech Republic
[10]Nuclear Physics Institute AS CR, 250 68 Řež/Prague, Czech Republic
[11]University of Frankfurt, Frankfurt, Germany
[12]Institute of Physics, Bhubaneswar 751005, India
[13]Indian Institute of Technology, Mumbai, India
[14]Indiana University, Bloomington, Indiana 47408, USA
[15]Alikhanov Institute for Theoretical and Experimental Physics, Moscow, Russia
[16]University of Jammu, Jammu 180001, India
[17]Joint Institute for Nuclear Research, Dubna, 141 980, Russia
[18]Kent State University, Kent, Ohio 44242, USA
[19]University of Kentucky, Lexington, Kentucky, 40506-0055, USA
[20]Institute of Modern Physics, Lanzhou, China
[21]Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA
[22]Massachusetts Institute of Technology, Cambridge, MA 02139-4307, USA
[23]Max-Planck-Institut für Physik, Munich, Germany
[24]Michigan State University, East Lansing, Michigan 48824, USA
[25]Moscow Engineering Physics Institute, Moscow Russia
[26]NIKHEF and Utrecht University, Amsterdam, The Netherlands
[27]Ohio State University, Columbus, Ohio 43210, USA
[28]Old Dominion University, Norfolk, VA, 23529, USA
[29]Panjab University, Chandigarh 160014, India
[30]Pennsylvania State University, University Park, Pennsylvania 16802, USA
[31]Institute of High Energy Physics, Protvino, Russia
[32]Purdue University, West Lafayette, Indiana 47907, USA
[33]Pusan National University, Pusan, Republic of Korea
[34]University of Rajasthan, Jaipur 302004, India
[35]Rice University, Houston, Texas 77251, USA
[36]Universidade de Sao Paulo, Sao Paulo, Brazil
[37]University of Science & Technology of China, Hefei 230026, China
[38]Shandong University, Jinan, Shandong 250100, China
[39]Shanghai Institute of Applied Physics, Shanghai 201800, China
[40]SUBATECH, Nantes, France
[41]Texas A&M University, College Station, Texas 77843, USA
[42]University of Texas, Austin, Texas 78712, USA
[43]Tsinghua University, Beijing 100084, China
[44]United States Naval Academy, Annapolis, MD 21402, USA
[45]Valparaiso University, Valparaiso, Indiana 46383, USA
[46]Variable Energy Cyclotron Centre, Kolkata 700064, India
[47]Warsaw University of Technology, Warsaw, Poland
[48]University of Washington, Seattle, Washington 98195, USA
[49]Wayne State University, Detroit, Michigan 48201, USA
[50]Institute of Particle Physics, CCNU (HZNU), Wuhan 430079, China
[51]Yale University, New Haven, Connecticut 06520, USA
[52]University of Zagreb, Zagreb, HR-10002, Croatia

(Dated: November 10, 2010)

Two-particle azimuthal ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) correlations using a trigger particle with large transverse momentum (p_T) in $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}}$ =62 and 200 GeV from the STAR experiment at RHIC are presented. The near-side correlation is separated into a jet-like component, narrow in both $\Delta\phi$ and $\Delta\eta$, and the ridge, narrow in $\Delta\phi$ but broad in $\Delta\eta$. Both components are studied as a function of collision centrality, and the jet-like correlation is studied as a function of p_T. The behavior of the jet-like component is remarkably consistent for different collision systems, comparisons to PYTHIA simulations suggests it is produced by fragmentation. The width of the jet-like correlation is found to increase with the collision system size, a possible indication of modifications of the jet-like correlation through mechanisms such as gluon brehmstrahlung. This is the first observation of the ridge in $Cu + Cu$ collisions and in collisions at

$\sqrt{s_{NN}}$ = 62 GeV. The ridge is found to be substantially smaller at $\sqrt{s_{NN}}$ = 62 GeV than at $\sqrt{s_{NN}}$ = 200 GeV for the same number of participants (N_{part}). This is potentially a powerful measurement to test models used to describe production mechanisms for the ridge and the jet modifications.

PACS numbers: 25.75.-q,21.65.Qr,24.85.+p,25.75.Bh

I. INTRODUCTION

Jets are a useful probe of the hot, dense medium created in heavy-ion collisions at the Relativistic Heavy Ion Collider (RHIC) at Brookhaven National Laboratory. Jet quenching [1] was first observed as the suppression of nuclear modification factors in central Au+Au collisions with respect to p+p collisions for charged particles with large transverse momentum (p_T) [2–7]. Properties of jets at RHIC have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum (p_T).

Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed their significant modification in $Au + Au$ relative to $p + p$ and $d + Au$ collisions at top RHIC energy of $\sqrt{s_{NN}}$ = 200 GeV. For low $p_T^{associated}$, the amplitude of the away-side peak is greater and the shape is modified in $Au + Au$ [8, 9]. At intermediate p_T ($4 < p_T^{trigger} < 6$ GeV/c, $2 < p_T^{associated} < 4$ GeV/c), the away-side correlation peak is strongly suppressed [10]. At higher p_T, the away-side peak reappears without shape modification, but the away-side per trigger yield is smaller in $Au+Au$ collisions than in $p + p$ and $d + Au$ [11].

The associated particle distribution on near side of the trigger particle, the subject of this paper, is also significantly modified in central $Au + Au$ collisions, a phenomenon not predicted before RHIC data were available. In $p + p$ and $d + Au$ collisions, there is a peak narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. This peak is also present in $Au+Au$ collisons, but an additional structure which is narrow in azimuth but broad in pseudorapidity has been observed in central $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV [8, 12]. This structure, called the ridge, is independent of $\Delta\eta$ within the STAR acceptance, $|\Delta\eta| < 2.0$, within errors, and persists to high p_T ($p_T^{trigger} \approx 6$ GeV/c). While the inverse slope parameter of the spectrum of particles in the jet-like correlation increases with increasing $p_T^{trigger}$, the spectrum of particles in the ridge shows no dependence on $p_T^{trigger}$, and the inverse slope parameter is comparable to that of the inclusive hadron spectrum. Recent studies by PHOBOS of di-hadron correlations at lower transverse momenta ($p_T^{trig} > 2.5$ GeV/c, $p_T^{assoc} > 20$ MeV/c) show that the ridge is roughly independent of $\Delta\eta$ and extends over six units in $\Delta\eta$ [13]. A similar broad correlation in pseudorapidity is also evident in untriggered di-hadron correlations [14].

Several mechanisms for the production of the ridge have been proposed since the first observation of this new phenomenon. In one model [15] the ridge is proposed to be formed from gluon radiation emitted by a high-p_T parton propagating in medium with strong longitudinal flow. Another physics mechanism [16] explains the ridge as arising from the spontaneous formation of extended color fields in a longitudinally expanding medium due to the presence of plasma instabilities. It has been also suggested that the ridge is not actually caused by a hard parton but is an artifact of radial flow and the surface biased emission of the trigger particle, causing an apparent correlation between particles from the bulk and high-p_T trigger particles [17, 18]. Long-range pseudorapidity correlations formed in an initial state glasma have also been discussed as a mechanism for the ridge [19, 20]. The momentum-kick model proposes that the ridge forms as a fast parton traveling through the medium loses energy through collisions with partons in the medium, causing those partons to be correlated in space with the fast parton [21–23]. Parton recombination has also been proposed as a mechanism for the production of the ridge [24–26]. Recently, it has been suggested that triangular anisotropy in the initial collision geometry caused by event-by-event fluctuations can give rise to triangular flow which can have significant contributions to the ridge as well as to broad-away side correlation peak observed in heavy-ion collisions at RHIC [27–29].

In this paper we present new measurements of the system size and collision energy dependence of near-side di-hadron correlations in $Cu + Cu$ and $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV measured by the STAR experiment at RHIC. In particular, we investigate the centrality of the jet-like correlations and the ridge and transverse momentum dependence of the jet-like correlations. The properties of jet-like correlations in heavy-ion collisions are compared to those from d+Au collisions and PYTHIA simulations to look for possible medium modifications and broadening of the near-side jet-like correlation e.g. due to gluon bremsstrahlung [15]. While there are many models of the ridge available, few quantitative calculations exist that can be readily compared to the experimental data. The new results on the system size and energy dependence of the ridge yield presented in this article extend our knowledge of this phenomenon and, in combination with other measurements at RHIC, provide important quantitative input and constraints to model calculations.

II. EXPERIMENTAL SETUP AND DATA SAMPLE

The results presented below are based on data measured by the STAR experiment from $d+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV in 2003, $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}} = 200$ GeV in 2004, and $Cu+Cu$ collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV in 2005. The d+Au events were selected using a minimum bias (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located at 18 m from the nominal interaction point in the Au beam direction, accepting 95±3% of the d+Au hadronic cross section [30]. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) at forward rapidity ($3.3 < |\eta| < 5.0$) and the ZDCs. The minimum bias trigger for Au+Au collisions at $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV was obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions.

For Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV, an additional online trigger for central collisions was used. This trigger was based on a maximum energy deposited in the two ZDCs in combination with a minimum multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section.

In order to achieve a more uniform detector acceptance, only those events with the primary vertex position along the longitudinal beam direction (v_z) within 30 cm of the nominal collision point were used for the analysis. For the $d+Au$ this was expanded to $|v_z| < 50$ cm.

The STAR Time Projection Chamber (TPC) [31] was used for tracking of charged particles. The collision centrality was determined from the uncorrected number of charged tracks at mid-rapidity ($|\eta| < 0.5$) in the TPC. The charged tracks used for the centrality determination had a 3-D distance of closest approach (DCA) to the primary vertex of less than 3 cm and at least 10 fit points from the TPC. Each data sample was then divided into several centrality bins, and the fraction of the geometric cross section, the number of participating nucleons (N_{part}) and the number of binary collisions (N_{coll}) were calculated using Glauber Monte-Carlo model calculations [32].

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	$\sqrt{s_{NN}}$ [GeV]	No. of events [10^6]
Cu+Cu	0-60%	62	24
Au+Au	0-80%	62	8
d+Au	0-95%	200	3
Cu+Cu	0-60%	200	38
Au+Au	0-80%	200	28
Au+Au	0-12%	200	17

III. DATA ANALYSIS

A. Correlation technique

For the analysis presented in this paper we used charged particle tracks reconstructed in the TPC. Tracks were required to have at least 15 hit points in the TPC, a DCA to the primary vertex of less than 1 cm, and a pseudorapidity $|\eta| < 1.0$. As in previous di-hadron correlation studies in the STAR experiment [12, 33, 34], a high-p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$), $d^2N_{raw}/d\Delta\phi\, d\Delta\eta$, normalized by the number of trigger particles, $N_{trigger}$, and corrected for the efficiency/acceptance of associated tracks ϵ is studied:

$$\frac{d^2N}{d\Delta\phi\, d\Delta\eta}(\Delta\phi,\Delta\eta) = \frac{1}{N_{trigger}}\frac{1}{\epsilon(\phi,\eta,\Delta\phi,\Delta\eta)}\frac{d^2N_{raw}}{d\Delta\phi\, d\Delta\eta}, \tag{1}$$

The efficiency correction includes single charged track reconstruction efficiency, track merging and finite TPC pair acceptance in $\Delta\phi$ and $\Delta\eta$ as described in detail below. All results shown use $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$ unless otherwise noted.

B. Single charged track efficiency correction

The correlations are first corrected for the single charged track reconstruction efficiency in the TPC. This efficiency is determined by simulating the detector response to a charged particle and embedding these signals into a real event. This hybrid event is analyzed using the same software as for the real events. The efficiency for detecting a single track as a function of p_T, η, and centrality is determined from the number of simulated particles which were successfully reconstructed. The single track efficiency is approximately constant for $p_T > 2$ GeV/c and ranges from around 75% for central $Au+Au$ events to around 85% for peripheral $Cu+Cu$ events as shown in Figure 1. The efficiency for reconstructing a track in $d+Au$ is 89%. In the correlations each track pair is corrected for the efficiency for reconstructing the associated particle. Since the correlations are normalized by the number of trigger particles, no correction for the efficiency of the trigger particle is necessary.

C. Correction of track merging and track crossing effects in the TPC

The reconstruction of charged tracks from TPC hits is performed sequentially, with hits removed from the event as they were assigned to a track. If two tracks have small

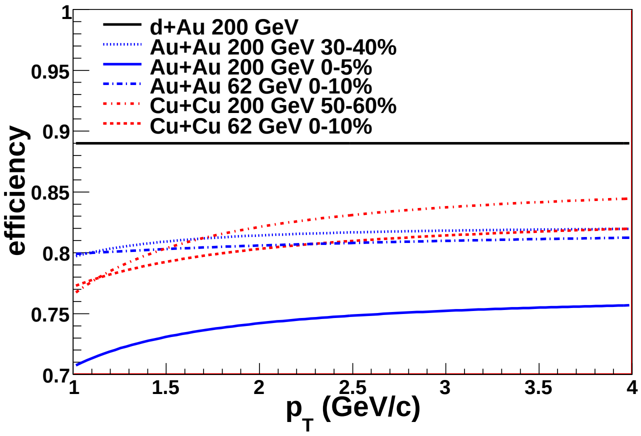

FIG. 1: Transverse momentum dependence of the reconstruction efficiency of charged particles in the TPC in various collision systems, energies and centrality bins.

angular separation in both pseudorapidity and azimuth, they are more difficult reconstruct because hits from each particle may not be resolved by the TPC. Therefore, if the particles have nearly identical momenta, their tracks are not distinguished. This effect, called track merging, reduces the number of pairs observed at small opening angles and results in an artificial dip in the raw correlations centered at $(\Delta\phi, \Delta\eta) = (0{,}0)$.

Figure 2 shows an example of $(\Delta\phi,\ \Delta\eta)$ two-particle di-hadron correlation function in central Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV, along with the corresponding $\Delta\phi$ and $\Delta\eta$ projections for $p_T^{trig} = 3$-6 GeV/c and $p_T^{assoc} > 1.5$ GeV/c. For small angular separations, a clear dip in the raw correlations is visible and must be corrected for in order to extract the yield of associated particles on the near side.

Another similar effect is evident in the active TPC volume from high-p_T tracks which cross. While hits are not lost in this case, one track may lose hits near the crossing point and therefore be split into two shorter tracks. Shorter tracks are less likely to meet the track selection criteria. This effect causes four dips near $(\Delta\phi, \Delta\eta) = (0{,}0)$ but slightly displaced in $\Delta\phi$. The location and width of these dips in $\Delta\phi$ is dependent on the relative helicities and the p_T intervals of the trigger and associated particles. The helicity h is given by

$$h = \frac{-qB}{|qB|} \tag{2}$$

where q is the charge of the particle and B is the magnetic field. Figure 3 displays the correlation function from Figure 2 in four different helicity combinations of trigger and associated particles demonstrating the existence of a finer substructure of the dip at the near side. When the helicities of the trigger and associated particles are the same, the percentage of overlapping hits is greater. Because the track curvature is a steeply falling function of the track p_T, it is more likely for higher p_T associated particles to be merged with a trigger particle than lower p_T particles. However, track pairs are lost whether the pair is part of the combinatorial background or part of the signal. This effect means that the magnitude of the dip is greater in central collisions where the background is greater and decreases with increasing p_T because of the decreasing background.

One way to correct for the track merging effect is to remove pairs from mixed events that would have merged in real data. The environment in mixed events must be similar to real data in order for pair rejection to be accurately reproduced. The reference multiplicities of both events were required to be within 10 of each other to assure similar track density. In addition, mixed events were required to have vertices within 2 cm of each other along the beam axis in order to ensure similar geometric acceptance, and avoid a different dip shape in $\Delta\eta$. In order to calculate accurately the percentage of merged hits, the origin of the associated track was shifted to the vertex of the event which the trigger particle originated from.

The fraction of merged hits was used as a threshold for merging. Previous analyses of low momenta tracks have shown that eliminating pairs from both data and mixed events with a fraction of merged hits greater than 10% was sufficient to correct for merging [35]. The calculation of the number of hits which may be shared by the tracks does not describe the performance of the TPC perfectly and some track pairs which were calculated to share more than 10% of their hits are reconstructed. These pairs are discarded even if they were reconstructed so that the percentage of merged track pairs is the same in the data and the mixed events.

The correlation function for a given helicity combination of trigger and associated particles was corrected by its corresponding mixed events. After this correction, a small residual dip remains because mixed events do not recreate the dip perfectly. While the mixed events correct for the dip due to true track merging well, they do not correct for track crossing as well. The remaining dip is then corrected for using the symmetry of the correlations. Since the data are symmetric about $\Delta\phi = 0$, the data on the same side as the dip are discarded and replaced by the data on the side without the dip. Then the data are reflected about $\Delta\eta = 0$ and added to the unreflected data to minimize statistical fluctuations. This method is only applied to $|\Delta\phi| < 1.05$ and $|\Delta\eta| < 0.67$ because it is computationally intensive and track merging only affects small angular separations. For large $\Delta\phi$ and $\Delta\eta$, the method described in the next section is applied. The track merging correction is done for each di-hadron correlation function separately with the appropriate cuts on $p_T^{trigger}$, $p_T^{associated}$ and collision centrality. An example of the di-hadron correlation function before and after the track merging correction is demonstrated in Figure 4.

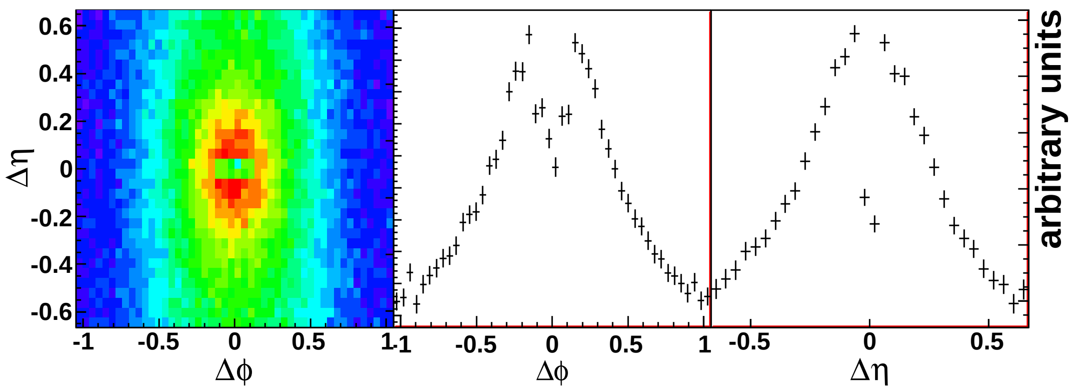

FIG. 2: Two-particle di-hadron correlation function in ($\Delta\phi$, $\Delta\eta$) space for p_T^{trig}=3-6 GeV/c and $p_T^{assoc} > 1.5$ GeV/c in central Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV, not corrected for track merging (left). The projection of the correlation function in $\Delta\phi$ for $|\Delta\eta| < 0.042$ is shown in the middle and the projection in $\Delta\eta$ for $|\Delta\phi| < 0.17$ on the right.

System	σ/σ_{geo} [%]	N_{part}	N_{coll}	$b_{\Delta\phi}$	$\langle v_2^{trigger}\rangle$ [%]	$\langle v_2^{assoc}\rangle$ [%]
d+Au 200 GeV	0-80	$8.3^{+0.4}_{-0.4}$	$7.5^{+0.4}_{-0.4}$	0.158±0.006	0.0	0.0
Cu+Cu 62 GeV	0-10	96^{+1}_{-3}	162^{+12}_{-14}	0.889±0.010	$9.9^{+0.5}_{-1.8}$	$8.7^{+0.5}_{-1.8}$
	10-30	64^{+1}_{-1}	$87.9^{+7.9}_{-7.9}$	0.515±0.006	$11.6^{+0.5}_{-1.2}$	$10.6^{+0.2}_{-0.6}$
	30-60	$25.7^{+0.4}_{-0.6}$	$27.6^{+1.2}_{-1.6}$	0.201±0.005	$14.5^{+1.0}_{-2.0}$	$11.9^{+0.2}_{-1.0}$
Au+Au 62 GeV	0-10	320.3		3.582±0.019	$8.5^{+0.3}_{-3.2}$	$7.6^{+0.3}_{-1.9}$
	10-40	168.0		1.846±0.010	$17.5^{+0.3}_{-3.2}$	$15.0^{+0.3}_{-1.9}$
	40-80	39.4		0.446±0.009	$21.2^{+0.3}_{-5.9}$	$18.6^{+0.3}_{-3.4}$
Cu+Cu 200 GeV	0-10	99^{+1}_{-1}	189^{+15}_{-13}	1.759±0.007	$8.0^{+0.7}_{-1.1}$	$8.8^{+0.1}_{-0.5}$
	10-20	75^{+1}_{-1}	$123.6^{+9.4}_{-8.3}$	1.206±0.006	$10.5^{+0.6}_{-1.1}$	$11.3^{+0.0}_{-0.6}$
	20-30	54^{+1}_{-1}	$77.6^{+5.4}_{-4.7}$	0.815±0.006	$11.2^{+0.7}_{-1.3}$	$12.5^{+0.1}_{-0.7}$
	30-40	$38^{+1.0}_{-1.0}$	$47.7^{+2.8}_{-2.7}$	0.537±0.006	$10.7^{+0.9}_{-1.5}$	$13.0^{+0.1}_{-0.8}$
	40-50	$26.2^{+0.5}_{-0.4}$	$29.2^{+1.6}_{-1.4}$	0.337±0.005	$11.2^{+2.2}_{-3.2}$	$12.6^{+0.5}_{-1.8}$
	50-60	$17.2^{+0.4}_{-0.2}$	$16.82^{+0.86}_{-0.69}$	0.209±0.005	$10.3^{+0.6}_{-1.5}$	$12.3^{+0.0}_{-1.3}$
Au+Au 200 GeV	0-12	316^{+6}_{-5}	900^{+71}_{-64}	13.255±0.003	$8.5^{+2.1}_{-2.1}$	$7.3^{+1.5}_{-1.5}$
	10-20	235^{+8}_{-9}	591^{+52}_{-60}	4.683±0.007	$14.6^{+2.0}_{-2.0}$	$13.0^{+1.2}_{-1.2}$
	20-30	167^{+9}_{-11}	369^{+41}_{-51}	3.222±0.006	$17.8^{+2.1}_{-2.1}$	$16.5^{+1.2}_{-1.2}$
	30-40	116^{+9}_{-11}	220^{+30}_{-38}	2.094±0.006	$18.9^{+2.3}_{-2.3}$	$18.1^{+1.4}_{-1.4}$
	40-50	77^{+9}_{-10}	123^{+23}_{-27}	1.267±0.006	$11.5^{+2.3}_{-2.3}$	$19.0^{+2.5}_{-2.5}$
	50-60	48^{+8}_{-10}	64^{+14}_{-19}	0.738±0.006	$10.1^{+2.3}_{-2.3}$	$17.6^{+2.8}_{-2.8}$
	60-70	27^{+6}_{-8}	$29.5^{+8.2}_{-11}$	0.386±0.006	$8.5^{+2.2}_{-2.2}$	$15.4^{+2.9}_{-2.9}$
	70-80	14^{+4}_{-5}	$12.3^{+4.4}_{-5.2}$	0.183±0.006	$6.6^{+1.9}_{-1.9}$	$12.8^{+2.8}_{-2.8}$

TABLE II: The background terms $b_{\Delta\phi}$ and elliptic flow values of trigger ($v_2^{trigger}$) and associated particles (v_2^{assoc}) for different enegy and collision systems as a function of centrality bins defined by the fraction of geometric cross section (σ/σ_{geo}), number of participants (N_{part}) and binary collisions (N_{coll}). The $p_T^{trigger}$ = 3-6 GeV/c and $p_T^{associated} > 1.5$ GeV/c selections were used.

D. Pair acceptance correction

With the restriction that each track falls within $|\eta| < 1.0$, there is a limited acceptance for track pairs. For $\Delta\eta \approx 0$, the geometric acceptance of the TPC for track pairs is $\approx 100\%$, however, near $\Delta\eta \approx 2$ the acceptance is close to 0%. In azimuth the acceptance is limited by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in azimuth. To correct for the geometric acceptance, the distribution of tracks as a function of η and ϕ was recorded for both trigger and associated particles. A random η and ϕ was chosen from each of these distributions to reconstruct a random $\Delta\eta$ and $\Delta\phi$. This was done for at least as many track pairs as in the data. This distribution was then normalized to give the geometrical acceptance correction for pairs.

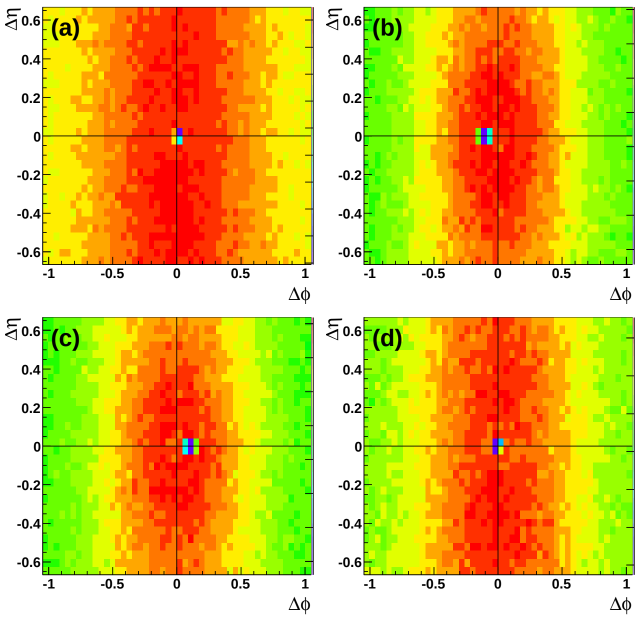

FIG. 3: The dip region in $(\Delta\phi,\ \Delta\eta)$ uncorrected charged-charged correlations for 3 GeV/c $< p_T^{trig} <$ 6 GeV/c and 1.5 GeV/c $< p_T^{assoc} < p_T^{trig}$ unfolded to four helicity combinations of trigger and associated particles: (a) (h_{trig}, h_{assoc}) = (1,1), (b) (h_{trig}, h_{assoc}) = (1,-1), (c) (h_{trig}, h_{assoc}) = (-1,1), and (d) (h_{trig}, h_{assoc}) = (-1,-1).

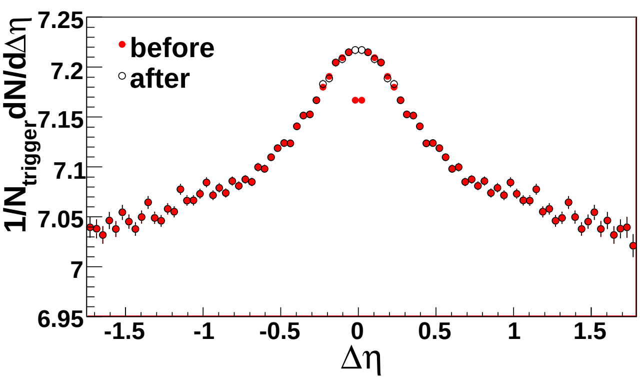

FIG. 4: The raw correlation in $\Delta\eta$ for charged-charged correlations for 3 GeV/c $< p_T^{trig} <$ 6 GeV/c and 1.5 GeV/c $< p_T^{assoc} < p_T^{trig}$ for 0-12% central Au+Au collisions before and after the track merging correction is applied

E. Subtraction of anisotropic elliptic flow background

Correlations of particles with the reaction plane due to the existence of anisotropic elliptic flow (v_2) in non-central heavy-ion collisions are indirectly reflected in di-hadron correlations and have to be subtracted for studies of the ridge. This background is approximated by

$$B_{\Delta\phi}[-a,a] \equiv b_{\Delta\phi} \int_{-a}^{a} d\Delta\phi \left(1 + 2\langle v_2^{trig}\rangle\langle v_2^{assoc}\rangle \cos 2\Delta\phi\right) \quad (3)$$

The level $b_{\Delta\phi}$ of the background is determined using the Zero Yield At Minimum (ZYAM) method [8]. In order to minimize the statistical bin-to-bin fluctuations the level of the background is taken as an average in the minimum bin and two adjacent bins.

The ZYAM method is commonly used for di-hadron correlations [11, 36, 37] at RHIC and is justified if the near- and away-side peaks are separated by a 'signal-free' region. At lower transverse momenta ($p_T < 2$ GeV/c) and in central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV the broadening of the away-side correlation peak may cause overlap of the near- and away-side peaks and consequently makes the ZYAM normalization procedure ambiguous. Alternatively, a decomposition of the correlation function using a fit function containing the anisotropic flow modulation of the combinatorial background and components describing the shape of the correlation peaks could be used instead [38, 39]. Despite these limitations we use the ZYAM prescription to remain consistent with our earlier measurements of the near-side ridge [12]. ZYAM used with conservative bounds on v_2 will, if anything, underestimate the ridge.

For all collision systems and energies studied, the uncertainty bounds on v_2 were determined by comparing different methods for the v_2 measurement. v_2 and systematic errors on v_2 for $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 62 GeV are from [40]. v_2 and systematic errors on v_2 for $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV are as described in [12]. For $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The upper bound on v_2 is from the reaction plane method using the forward TPCs for the determination of the reaction plane, the lower bound comes from the four particle cumulant method, and the average of the two is the nominal value. v_2 for $Cu+Cu$ collisions at $\sqrt{s_{NN}}$ = 62 GeV and 200 GeV is from [41]. The nominal value is given by v_2 from the reaction plane method using the forward TPCs for the determination of the reaction plane and the systematic error is 10% for 0-40% centrality and 20% for 40-60% centrality. In Table II is shown an example of $b_{\Delta\phi}$ values and v_2 of trigger and associated particles for $Cu+Cu$ and $Au+Au$ collisions at both studied collision energies. The centrality bins are characterized by the fraction of geometric cross section σ/σ_{geo}, number of participants N_{part} and number of binary collisions N_{coll}.

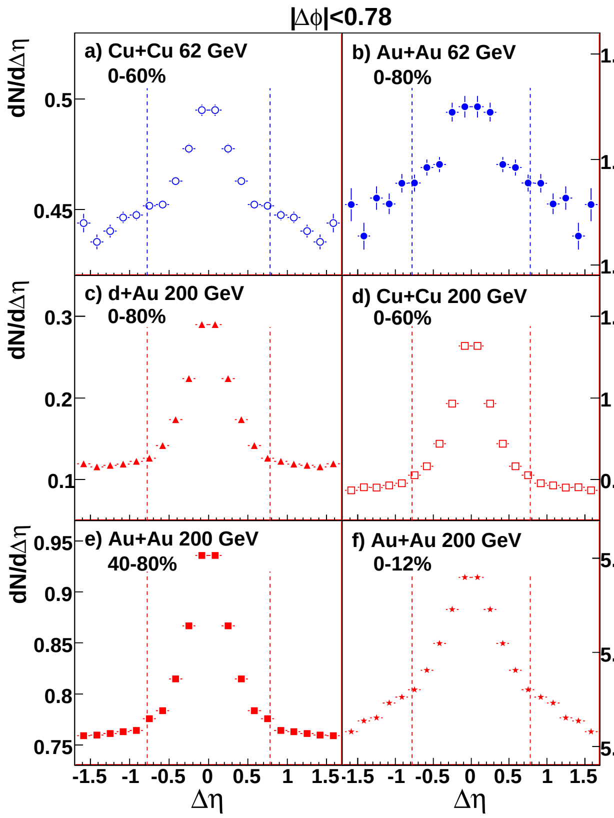

FIG. 5: Sample correlations in $\Delta\eta$ for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$ for (a) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au+Au$ at $\sqrt{s_{NN}} = 62$ GeV, (c) minimum bias d+Au at $\sqrt{s_{NN}} = 200$ GeV, (d) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}} = 200$ GeV, (e) 40-80% $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV, and (f) 0-12% central $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV. Lines show the $\Delta\eta$ range where the jet-like yield is determined. Color online.

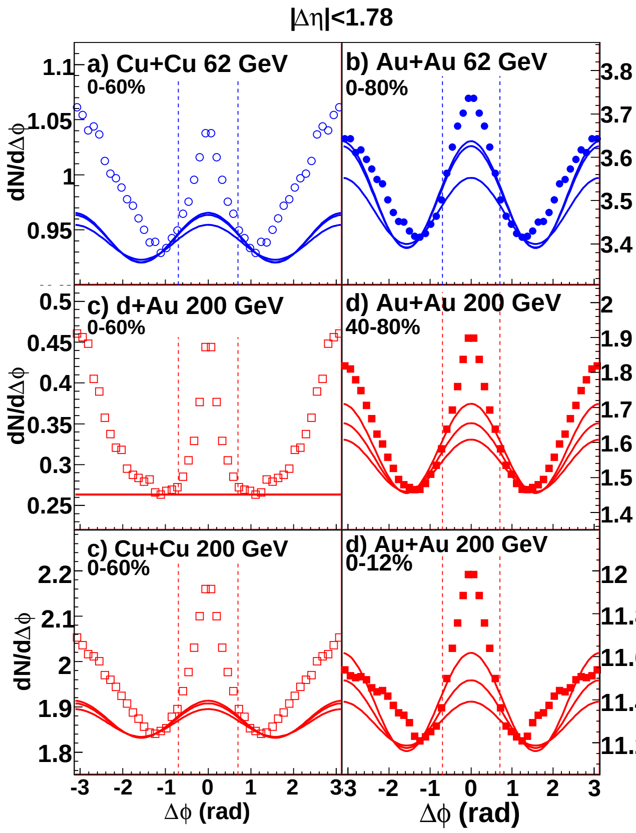

FIG. 6: Sample correlations in $\Delta\phi$ for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$ for (a) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}} = 62$ GeV (b) 0-80% central $Au+Au$ at $\sqrt{s_{NN}} = 62$ GeV, (c) minimum bias d+Au at $\sqrt{s_{NN}} = 200$ GeV, (d) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}} = 200$ GeV, (e) 40-80% $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV, and (f) 0-12% central $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV. Lines show the estimated background using the ZYAM method with the range of v_2 used for the determination of the systematic errors.

With both methods for subtracting the ridge contribution to Y_J described below, the systematic errors due to v_2 cancel out in the jet-like yield, assuming that v_2 is roughly independent of $\Delta\eta$. This assumption is based on the available data measured at mid-rapidity $|y| < 1$ [42, 43].

F. Yield extraction

To quantify the strength of the near-side side correlation it is assumed that it can be decomposed to a jet-like component, narrow in both azimuth and pseudorapidity, and a ridge component which is independent of $\Delta\eta$. This approach is consistent with the method in [12]. For the kinematic cuts applied to $p_T^{trigger}$ and $p_T^{associated}$, the jet-like correlation is contained within $|\Delta\eta| < 0.78$ and $|\Delta\phi| < 0.78$.

To study the jet-like correlation and the ridge quantitatively we adopt notation from [12], and introduce the projection of the di-hadron correlation function from Eq. (1) on the $\Delta\eta$ axis:

$$\left.\frac{dN}{d\Delta\eta}\right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}; \tag{4}$$

and similarly on the $\Delta\phi$ axis:

$$\left.\frac{dN}{d\Delta\phi}\right|_{a,b} \equiv \int_{|\Delta\eta|\in[a,b]} d\Delta\eta \frac{d^2N}{d\Delta\phi d\Delta\eta}. \tag{5}$$

To determine the jet-like yield of associated charged particles two methods are used. Under the assumption that the jet-like yield is confined within $|\Delta\eta| < 0.78$, subtracting the $\Delta\phi$ projections:

$$\frac{dN_J}{d\Delta\phi}(\Delta\phi) = \left.\frac{dN}{d\Delta\phi}\right|_{0,0.78} - \frac{0.78}{1.0}\left.\frac{dN}{d\Delta\phi}\right|_{0.78,1.78} \tag{6}$$

removes both the elliptic flow and ridge contributions. Since the second $\Delta\phi$ projection is calculated in a larger

$\Delta\eta$ window, it has to be scaled by a factor 0.78/1.0, the ratio of the $\Delta\eta$ width in the region containing the jet-like correlation and the ridge to the width of the region containing only the ridge. This subtracts both the ridge and v_2 simultaneously since both are independent of $\Delta\eta$.

The jet-like yield $Y_J^{\Delta\phi}$ is then obtained by integrating Eq.(6)

$$Y_J^{\Delta\phi} = \int_{-0.78}^{0.78} d\Delta\phi \, \frac{dN_J}{d\Delta\phi}(\Delta\phi) \, , \tag{7}$$

The second method for jet-like yield determination is based on the $\Delta\eta$ projection at the near-side:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - b_{\Delta\eta} \tag{8}$$

as v_2 is independent of pseudo-rapidity within the STAR acceptance. The background level $b_{\Delta\eta}$ is determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$. Integrating Eq. (8) over $\Delta\eta$ defines then the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \, \frac{dN_J}{d\Delta\eta}(\Delta\eta) \, , \tag{9}$$

To minimize the effects of statistical fluctuations in the determination of the background and maximizing the data statistics, the ridge yield Y_{ridge} is determined by first integrating Eq. (5) over the entire $\Delta\eta$ region then subtracting the modulated elliptic flow background $B_{\Delta\phi}$ and the jet-like contribution $Y_J^{\Delta\eta}$:

$$Y_{ridge} = \int_{-1.78}^{1.78} d\Delta\eta \, \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - B_{\Delta\phi}[-0.78, 0.78] - Y_J^{\Delta\eta,\Delta\phi}. \tag{10}$$

IV. RESULTS

A. Sample Correlations

Figure 5 shows fully corrected $\Delta\eta$ projections of sample correlations on the near side ($|\Delta\phi| < 0.78$) before background subtraction for $d + Au$, $Cu + Cu$ and $Au + Au$ collisions at energy $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV. The trigger particles were selected with transverse momentum $3 < p_T^{trigger} < 6$ GeV/c and the associated particles with 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$. The data show a clear jet-like peak sitting a top of the background. The level of the background is increasing with energy and system size as expected as more bulk particles are produced in the collision.

Examples of the complementary projections in $\Delta\phi$ before background subtraction for ($|\Delta\eta| < 1.78$) are shown in Figure 6. The elliptic flow modulated background is shown as curves. The middle curve corresponds to background calculated with the nominal value of v_2. The upper (lower) curve corresponds to the background if the upper (lower) bound on v_2 is used instead. Note that the uncertainty in the size of the elliptic modulated background affects only the ridge yield as the elliptic flow contribution to the jet-like yield in $\Delta\phi$ cancels out in Eq. (6).

The background subtracted $\Delta\eta$ (closed symbols) and $\Delta\phi$ (opened symbols) sample correlation functions at near side for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ from Figure 5 and Figure 6 are shown in Figure 7. For the given kinematic selection, the extracted jet-like correlation peaks in both $\Delta\eta$ and $\Delta\phi$ projections look very similar in all studied systems and collision energies. The jet-like correlation yields discussed through the rest of the paper are obtained from the $\Delta\eta$ projection method; the $\Delta\phi$ method is only used to for determining the width of the jet-like correlation in $\Delta\phi$. Below, the dependence of the near side jet-like correlation yield and Gaussian width of the jet-like correlation peak on collision centrality and the transverse momentum of the trigger and associated particles are studied in detail.

B. The near-side jet-like component

The centrality dependence for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$ of the jet-like correlation yield is plotted in Figure 8. The jet-like correlation yield at $\sqrt{s_{NN}} = 62$ GeV is lower than that at $\sqrt{s_{NN}} = 200$ GeV by about factor of three which can be understood as a result of a steeply falling jet spectrum folded with the fragmentation function. For a given number of participating nucleons, N_{part}, and collision energy, $\sqrt{s_{NN}}$, there is no significant difference between the $d + Au$, $Cu + Cu$ and $Au + Au$ collisions observed, as expected if the jet-like correlation were dominantly produced by vacuum fragmentation. The jet-like correlation yield shows a slight increase at higher N_{part}, indicating possible medium modification of the jet-like correlation in central $Au+Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. The measured yields are also compared to PYTHIA simulations shown as band in Figure 8. For these studies PYTHIA version 6.4.10 [44] tune A [45], which matches the data from the Tevatron at $\sqrt{s}$ =1.8 TeV and describes also well pion and proton inclusive spectra at RHIC energies [46, 47]. PYTHIA is somewhat above the data, even for $d + Au$ collisions. However, considering the fact that the data are from heavy-ion collisions and the PYTHIA simulations are for $p + p$ collisions, the agreement is surprisingly good.

Next, the dependence of the jet-like yield on $p_T^{trigger}$ for 1.5 GeV/$c < p_T^{associated} < p_T^{trigger}$ is plotted in Figure 9 for all studied collision systems and energies. Data

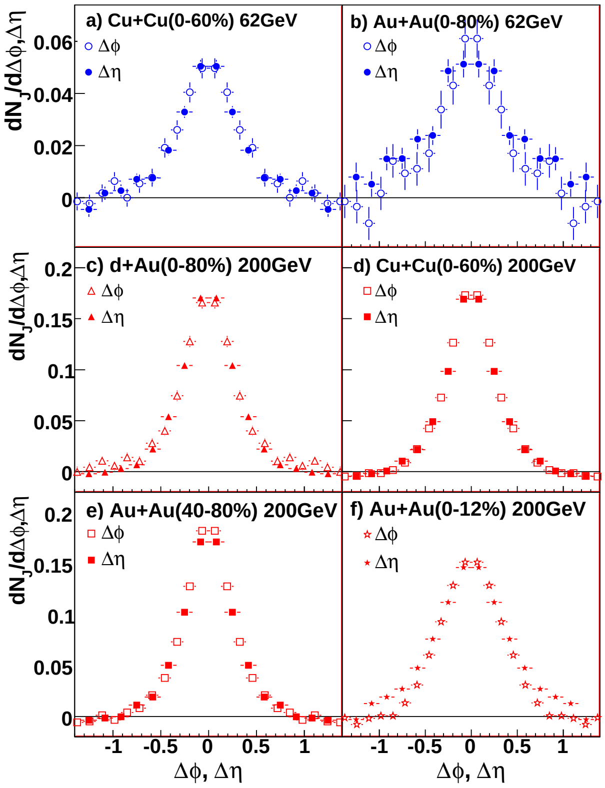

FIG. 7: Background subtracted sample correlations for $3 < p_T^{trigger} < 6$ GeV/c and 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ on the near-side for (a) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}}$ = 62 GeV, (b) 0-80% central $Au+Au$ at $\sqrt{s_{NN}}$ = 62 GeV, (c) minimum bias $d+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV, (d) 0-60% central $Cu+Cu$ at $\sqrt{s_{NN}}$ = 200 GeV, (e) 40-80% central $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV and (f) 0-12% central $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV. The dependence of the jet-like correlation is shown as a function of $\Delta\phi$ in open symbols and $\Delta\eta$ in closed symbols. Lines show the $\Delta\phi$ range where the jet-like yield is determined. Color online.

from $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV are shown separately for peripheral (40-80%) and central (0-12%) $Au+Au$ collisions to study the medium modification of the jet-like correlation with the collision centrality. The jet-like yield increases constantly with $p_T^{trigger}$ for both $Cu+Cu$ and $Au+Au$ and for $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV. The effect of a steeper jet spectrum at $\sqrt{s_{NN}}$ = 62 GeV relative to $\sqrt{s_{NN}}$ = 200 GeV discussed above is now reflected in the increasing difference between the jet-like correlation yields at the two energies. This difference in jet-like correlation yield between the two studied collision energies increases with $p_T^{trigger}$ from about a factor of two at $p_T^{trigger}$ = 2.5 GeV/c to a factor of four at $p_T^{trigger}$ = 5.5 GeV/c. No difference between $d+Au$, $Cu+Cu$ and peripheral $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV is observed, however at the same energy the jet-like yield is larger in central $Au+Au$ collisions than in

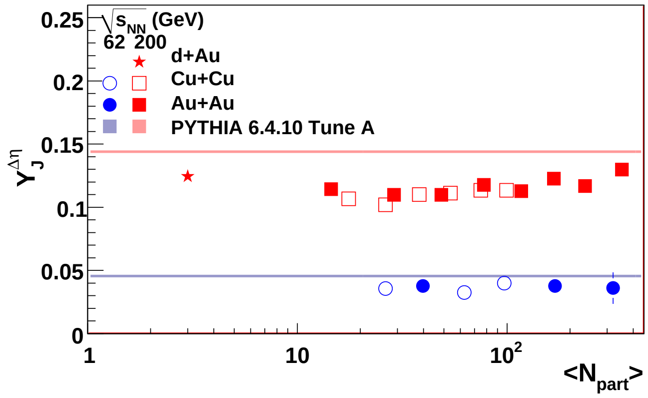

FIG. 8: Dependence of jet-like correlation yields on N_{part} for minimum bias $d+Au$, 0-60% $Cu+Cu$ at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV, 0-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 62 GeV, and 0-12% and 40-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV. Comparisons to PYTHIA, N_{part} =2, are shown as shaded bands. Color online.

peripheral collisions or $Cu+Cu$ collisions. This finding is consistent with the jet-like correlation component arising from fragmentation, and indicates possible medium modification of the jet-like yield in central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. Comparisons to PYTHIA simulation are shown as bands in Figure 9. In general, the jet-like correlation yield is described well by PYTHIA at larger $p_T^{trigger}$ ($p_T^{trigger} > 4$ GeV/c), while at lower $p_T^{trigger}$ values PYTHIA predicts larger jet-like yield than observed in the data, which is not surprising as in this p_T range physics processes such as parton coalescence and recombination [48–51] could contribute to particle production.

The spectra of particles associated with the jet-like correlation and their comparison to PYTHIA simulations for

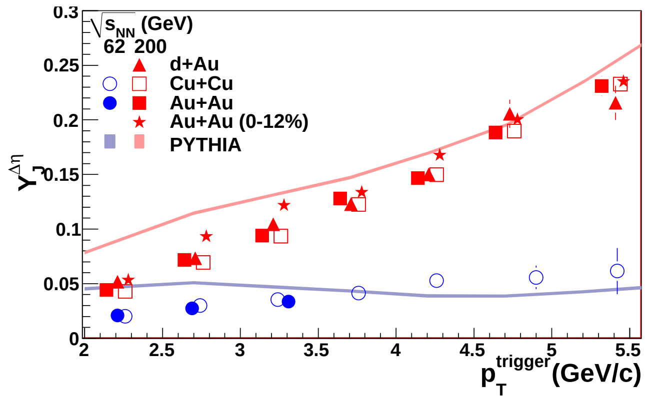

FIG. 9: Dependence of jet-like correlation yield on $p_T^{trigger}$ for minimum bias $d+Au$, 0-60% $Cu+Cu$ at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV, 0-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 62 GeV, and 0-12% and 40-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV. Comparisons to PYTHIA are shown as shaded bands. Color online.

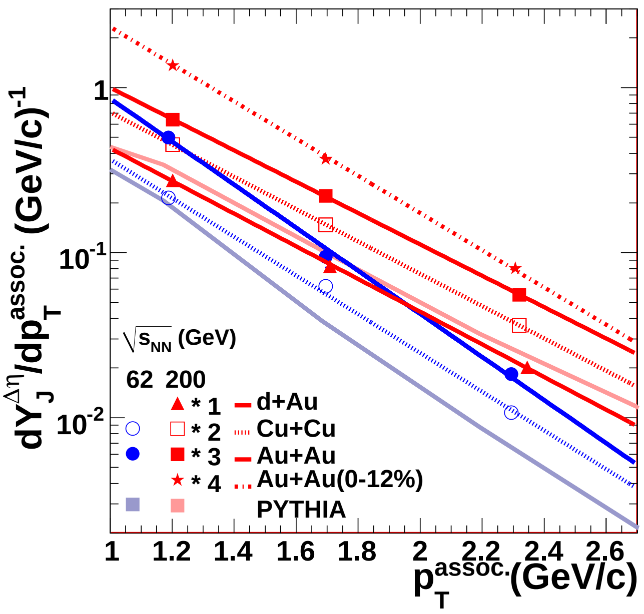

FIG. 10: Dependence of jet-like correlation yield on $p_T^{associated}$ for minimum bias $d+Au$, 0-60% $Cu+Cu$ at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV, 0-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 62 GeV, and 0-12% and 40-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV. Comparisons to PYTHIA are shown as shaded bands. Color online.

$3 < p_T^{trigger} < 6$ GeV/c are shown in Figure 10. For the same $p_T^{trigger}$ selection, the mean transverse momentum fraction z_T carried by the leading hadron is larger due to the steeper jet spectrum at $\sqrt{s_{NN}}$ = 62 GeV than at $\sqrt{s_{NN}}$ = 200 GeV. This is then reflected in softer $p_T^{associated}$ spectra at $\sqrt{s_{NN}}$ = 62 GeV. The inverse slope parameters from an exponential fit to these data are shown in Table III. There is no difference seen between $Cu+Cu$ and peripheral $Au+Au$ in either the data points or the extracted inverse slope parameter. The inverse slope parameter of the central $Au+Au$ data at $\sqrt{s_{NN}}$ = 200 GeV is somewhat lower than the other data at $\sqrt{s_{NN}}$ = 200 GeV, largely because of the larger yield at the lowest $p_T^{associated}$. This also indicates that there is some modification of the jet-like correlation at low p_T. Comparisons to PYTHIA demonstrate that PYTHIA does not describe the data well, particularly at lower p_T and at $\sqrt{s_{NN}}$ = 62 GeV. This is not surprising since PYTHIA is tuned better at higher p_T. The disagreement between the dependence of the jet-like yield on $p_T^{associated}$ in the data and in PYTHIA may explain why the data in Figure 9 are described better at $\sqrt{s_{NN}}$ = 200 GeV than $\sqrt{s_{NN}}$ = 62 GeV.

Figure 11 shows the Gaussian widths of the $\Delta\phi$ and $\Delta\eta$ projections of the near-side jet-like peak as a function of $p_T^{trigger}$, $p_T^{associated}$, and N_{part} along with PYTHIA simulations. There are no significant differences between collision systems except for central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. While no dependence on collision system is observed, there is a clear increase in the width with increasing N_{part} in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. As for the jet-like yields, this indicates that the jet-like correlation is modified in central $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. PYTHIA predicts that the jet-like correlation should be wider at $\sqrt{s_{NN}}$ = 62 GeV than at 200 GeV, but this is not seen in the data. PYTHIA also predicts a greater width in $\Delta\eta$ at the lowest $p_T^{associated}$ and this is not seen in the data.

Overall it can be concluded that the agreement between the different collision systems and energies shows remarkably little dependence of the jet-like per trigger yield on the system size. In contrast to the peripheral $Au+Au$ data, the central $Au+Au$ data show indications that the jet-like correlation is modified. The model in [53], a hypothesis for the formation of the ridge through gluon bremsstrahlung, does not produce a ridge broad enough to agree with the data, however, it is possible that a similar mechanism could explain the broadening of the jet-like correlation.

TABLE III: Inverse slope parameter k (MeV/c) of $p_T^{associated}$ for fits of data in Figure 10. The inverse slope parameter from a fit to π^- in $Au+Au$ from [52] above 1.0 GeV/c is k = 280.9 ± 0.4 MeV/c for $\sqrt{s_{NN}}$ = 62 GeV and is k = 330.9 ± 0.3 MeV/c for $\sqrt{s_{NN}}$ = 200 GeV. Statistical errors only.

	$\sqrt{s_{NN}}$ = 62 GeV	$\sqrt{s_{NN}}$ = 200 GeV
$Au+Au$	332 ± 13	457 ± 4 (40-80%) 384 ± 3 (0-12%)
$Cu+Cu$	370 ± 9	443 ± 3
$d+Au$		438 ± 9
PYTHIA	417 ± 9	491 ± 3

C. The near-side ridge

In [12], we reported the first observation of the ridge in $Au+Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV and studied its transverse momentum dependence. Here, an investigation of the ridge centrality as well as energy and system size dependence is performed. The dependence of the ridge yield on N_{part} for $Cu+Cu$ and $Au+Au$ collisions is shown in Figure 12 for both energies studied. Contrary to the jet-like correlation yield, which shows no dependence on centrality, the ridge yield increases steeply with N_{part}. Within errors, there is no difference in ridge yield between $Cu+Cu$ and $Au+Au$ collisions at the same N_{part} demonstrating system independence of the ridge yield.

Quantitative comparisons to models are currently possible only with the momentum kick model [21]. The momentum kick model with the same kinematic selection criteria applied to charged particles describes quantitatively well the increase of the ridge yield with centrality. For a given collision energy the ridge yield for $Cu+Cu$ collisions is predicted to follow approximately that for

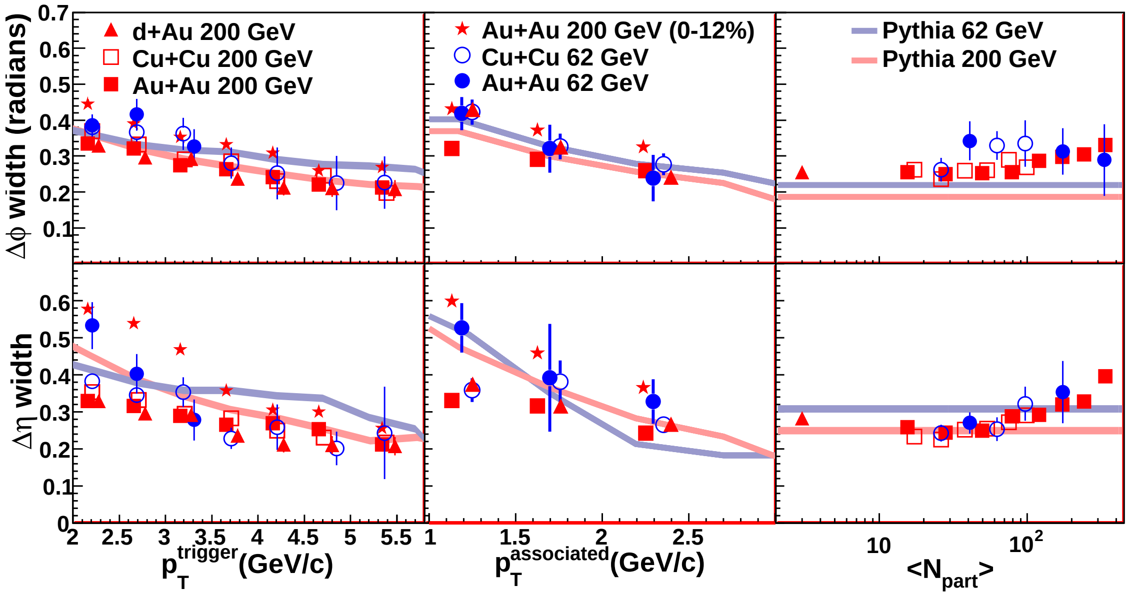

FIG. 11: Dependence of the widths in $\Delta\phi$ and $\Delta\eta$ on $p_T^{trigger}$, $p_T^{associated}$, and N_{part} for minimum bias $d+Au$, 0-60% $Cu+Cu$ at $\sqrt{s_{NN}}$ = 62 GeV and $\sqrt{s_{NN}}$ = 200 GeV, 0-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 62 GeV, and 0-12% and 40-80% $Au+Au$ at $\sqrt{s_{NN}}$ = 200 GeV. Comparisons to PYTHIA are shown as shaded bands. Color online.

$Au+Au$ collisions when plotted in terms of the number of participants in agreement with our observations. It is also possible that the recently suggested triangular flow [27] may contribute to the observed ridge yield.

Next, the energy dependence of the ridge yield is discussed which is potentially sensitive test of ridge models. Comparing the two collision energies studied, the ridge yield is observed to be considerably smaller at $\sqrt{s_{NN}}$ = 62 GeV than at $\sqrt{s_{NN}}$ = 200 GeV. Similar behavior was also observed for the jet-like correlation yield. Therefore a closer investigation of the centrality dependence of the ratio Y_{ridge}/Y_{jet} is reported in Figure 13. The ratio of the yields is independent of $\sqrt{s_{NN}}$ within errors. For the same kinematic selections, the data at $\sqrt{s_{NN}}$ = 62 GeV correspond to a lower jet energy implying the decrease of the ridge yield with energy as for the jet-like correlation yield.

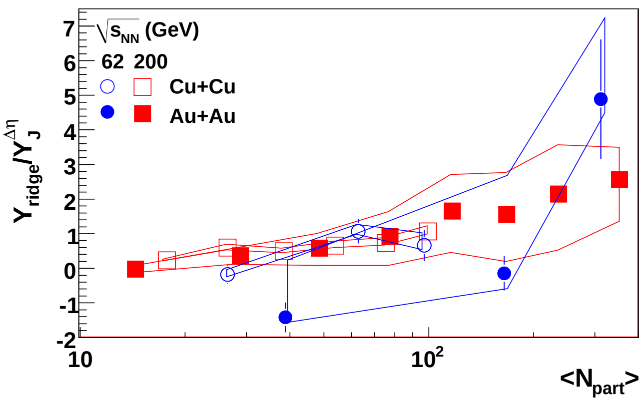

FIG. 13: Ratio of ridge and jet-like yields as a function of N_{part}. Color online.

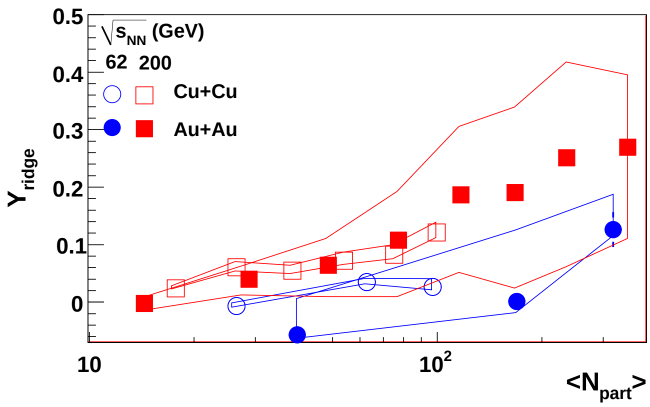

FIG. 12: Dependence of ridge yield on N_{part} for Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}}$= 62 and 200 GeV. Color online.

A direct quantitative comparison is currently possible only with the momentum kick model. For the same nucleus-nucleus collision at different energies, the ridge yield in the momentum kick model is predicted to scale approximately with the number of medium partons produced per participant which increases with increasing collision energy as $(ln\sqrt{s})^2$ [54]. For the collision energy range investigated here the predicted ratio is about 1.5 and agrees within errors with our measurement. It is very important to see predictions from the triangular flow model as well as other ridge models how the strength of triangular flow/ridge yield evolves with collision energy.

V. CONCLUSIONS

There is remarkably little dependence of the jet-like correlation on the collision system at both $\sqrt{s_{NN}} = 62$ GeV and $\sqrt{s_{NN}} = 200$ GeV except for central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV. In central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV the jet-like correlation is substantially broader than in peripheral collisions and collisions from other systems. This may indicate that fragmentation is modified in these collisions so that the parton fragments softer, perhaps due to a mechanism such as gluon bremsstrahlung. The reasonable agreement with PYTHIA is surprising, especially considering that PYTHIA is a $p + p$ event generator. This indicates that the near-side jet-like correlation is dominantly produced by vacuum fragmentation and that many features of this fragmentation are understood well quantitatively.

The ridge is observed not only in $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV as we reported earlier, but also in $Cu+Cu$ collisions and in both studied collision systems at lower energy of $\sqrt{s_{NN}} = 62$ GeV. This demonstrates that the ridge is not a feature unique to $Au + Au$ collisions at the top RHIC energy. The fact that the measured ridge/jet ratio is within errors equal for the two studied collision energies sets significant limits to models in which the ridge origin is not connected to jet. Combination of these data with future measurements at lower RHIC energies ($\sqrt{s_{NN}}$ = 7-39 GeV) as well as studies at the LHC will therefore be a powerful tool for the distinction between various theoretical models.

Acknowledgments

We thank the RHIC Operations Group and RCF at BNL, the NERSC Center at LBNL and the Open Science Grid consortium for providing resources and support. This work was supported in part by the Offices of NP and HEP within the U.S. DOE Office of Science, the U.S. NSF, the Sloan Foundation, the DFG cluster of excellence 'Origin and Structure of the Universe'of Germany, CNRS/IN2P3, FAPESP CNPq of Brazil, Ministry of Ed. and Sci. of the Russian Federation, NNSFC, CAS, MoST, and MoE of China, MSMT of the Czech Republic, FOM and NWO of the Netherlands, DAE, DST, and CSIR of India, Polish Ministry of Sci. and Higher Ed., Korea Research Foundation, Ministry of Sci., Ed. and Sports of the Rep. Of Croatia, and RosAtom of Russia.

---

[1] X.-N. Wang and M. Gyulassy, Phys. Rev. Lett. **68**, 1480 (1992).
[2] C. Adler et al. (STAR), Phys. Rev. Lett. **89**, 202301 (2002), nucl-ex/0206011.
[3] K. Adcox et al. (PHENIX), Phys. Rev. Lett. **88**, 022301 (2002), nucl-ex/0109003.
[4] K. Adcox et al. (PHENIX), Phys. Lett. **B561**, 82 (2003), nucl-ex/0207009.
[5] S. S. Adler et al. (PHENIX), Phys. Rev. Lett. **91**, 072301 (2003), nucl-ex/0304022.
[6] B. B. Back et al. (PHOBOS), Phys. Lett. **B578**, 297 (2004), nucl-ex/0302015.
[7] J. Adams et al. (STAR), Phys. Rev. Lett. **91**, 172302 (2003), nucl-ex/0305015.
[8] J. Adams et al. (STAR), Phys. Rev. Lett. **95**, 152301 (2005), nucl-ex/0501016.
[9] B. I. Abelev et al. (STAR) (2008), 0805.0622.
[10] C. Adler et al. (STAR), Phys. Rev. Lett. **90**, 082302 (2003), nucl-ex/0210033.
[11] J. Adams et al. (STAR), Phys. Rev. Lett. **97**, 162301 (2006), nucl-ex/0604018.
[12] B. I. Abelev et al. (STAR), Phys. Rev. **C80**, 064912 (2009), 0909.0191.
[13] B. Alver et al. (PHOBOS), Phys. Rev. Lett. **104**, 062301 (2010), 0903.2811.
[14] J. Adams et al. (STAR), Phys. Rev. **C73**, 064907 (2006), nucl-ex/0411003.
[15] N. Armesto, C. A. Salgado, and U. A. Wiedemann, Phys. Rev. Lett. **93**, 242301 (2004), hep-ph/0405301.
[16] A. Majumder, B. Muller, and S. A. Bass, Phys. Rev. Lett. **99**, 042301 (2007), hep-ph/0611135.
[17] S. A. Voloshin, Phys. Lett. **B632**, 490 (2006), nucl-th/0312065.
[18] C. A. Pruneau, S. Gavin, and S. A. Voloshin, Nucl. Phys. **A802**, 107 (2008), 0711.1991.
[19] A. Dumitru, F. Gelis, L. McLerran, and R. Venugopalan, Nucl. Phys. **A810**, 91 (2008), 0804.3858.
[20] F. Gelis, T. Lappi, and R. Venugopalan (2008), 0810.4829.
[21] C.-Y. Wong, Phys. Rev. **C78**, 064905 (2008), 0806.2154.
[22] C.-Y. Wong (2007), 0712.3282.
[23] C.-Y. Wong, Phys. Rev. **C76**, 054908 (2007), 0707.2385.
[24] R. C. Hwa and C. B. Yang (2008), 0812.2205.
[25] R. C. Hwa and C. B. Yang (2008), 0801.2183.
[26] C. B. Chiu and R. C. Hwa (2008), 0809.3018.
[27] B. Alver and G. Roland, Phys. Rev. **C81**, 054905 (2010), 1003.0194.
[28] B. H. Alver, C. Gombeaud, M. Luzum, and J.-Y. Ollitrault (2010), 1007.5469.
[29] H. Holopainen, H. Niemi, and K. J. Eskola (2010), 1007.0368.
[30] J. Adams et al. (STAR), Phys. Rev. Lett. **91**, 072304 (2003), nucl-ex/0306024.
[31] K. H. Ackermann et al., Nucl. Instrum. Meth. A **499**, 624 (2003).
[32] M. L. Miller, K. Reygers, S. J. Sanders, and P. Steinberg, Ann. Rev. Nucl. Part. Sci. **57**, 205 (2007), nucl-ex/0701025.
[33] B. I. Abelev et al. (STAR), Phys. Lett. **B683**, 123 (2010), 0904.1722.
[34] M. M. Aggarwal et al. (STAR), Phys. Rev. **C82**, 024912 (2010), 1004.2377.

[35] J. Adams et al. (STAR), Phys. Rev. **C71**, 044906 (2005), nucl-ex/0411036.

[36] J. Adams et al. (STAR), Phys. Rev. Lett. **95**, 152301 (2005), nucl-ex/0501016.

[37] A. Adare et al. (PHENIX), Phys. Rev. **C78**, 014901 (2008), 0801.4545.

[38] J. Adams et al. (Star), Phys. Rev. **C75**, 034901 (2007), nucl-ex/0607003.

[39] T. A. Trainor, Phys. Rev. **C81**, 014905 (2010), 0904.1733.

[40] B. I. Abelev et al. (the STAR), Phys. Rev. **C75**, 054906 (2007), nucl-ex/0701010.

[41] B. I. Abelev et al. (The STAR), Phys. Rev. **C81**, 044902 (2010), 1001.5052.

[42] B. B. Back et al. (PHOBOS), Phys. Rev. **C72**, 051901 (2005), nucl-ex/0407012.

[43] B. B. Back et al. (PHOBOS), Phys. Rev. Lett. **94**, 122303 (2005), nucl-ex/0406021.

[44] T. Sjostrand, S. Mrenna, and P. Skands, JHEP **05**, 026 (2006), hep-ph/0603175.

[45] R. D. Field (CDF) (2002), hep-ph/0201192.

[46] J. Adams et al. (STAR), Phys. Rev. **D74**, 032006 (2006), nucl-ex/0606028.

[47] J. Adams et al. (STAR), Phys. Lett. **B637**, 161 (2006), nucl-ex/0601033.

[48] R. J. Fries, B. Muller, C. Nonaka, and S. A. Bass, Phys. Rev. **C68**, 044902 (2003).

[49] V. Greco, C. M. Ko, and P. Levai, Phys. Rev. **C68**, 034904 (2003).

[50] V. Greco, C. M. Ko, and P. Levai, Phys. Rev. Lett. **90**, 202302 (2003).

[51] R. C. Hwa and C. B. Yang, Phys. Rev. **C67**, 034902 (2003).

[52] B. I. Abelev et al. (STAR), Phys. Lett. **B655**, 104 (2007), nucl-ex/0703040.

[53] N. Armesto, C. A. Salgado, and U. A. Wiedemann, Phys. Rev. **C72**, 064910 (2005), hep-ph/0411341.

[54] W. Busza, Acta Phys. Polon. **B35**, 2873 (2004), nucl-ex/0410035.